

Controlling Desktop Applications via the Mobile



This article introduces the reader to SignalR, which is a popular library for developing ASP.NET based solutions that have real-time connectivity functions. It provides an introduction to how desktop applications, digital signage and touchscreen kiosks can be controlled by mobile applications using SignalR as a real-time connectivity platform.

We are living in a world of instant communication. High-speed Internet, powerful but low-cost mobile devices and minute sensors combine to enable this new speed. Real-time connectivity can lead to real-time connections, whereby information is exchanged between users instantly or with negligible latency.

Real-time Web technologies have been around for over 10 years now, and popular platforms such as Facebook, Twitter and Google+ for communication are built on these technologies. We use such technologies while watching news updates or live matches, scores, etc. The best example of real-time connected applications is the chat application, where a client can broadcast messages to all other connected clients.

As the need and popularity for real-time connectivity has grown, many technology platforms have evolved, which use different approaches. SignalR is a popular library for developing ASP.NET based solutions that have real-time connectivity functions. In this article, we will look at how desktop applications, digital signage and touchscreen kiosks can be controlled by mobile apps using SignalR as a real-time connectivity platform. The article will also provide a basic feature comparison with some other popular products in the market.

Typical communication in a system is to push information from client to server, and also to get information from the server, on demand. It was a difficult task for the server to reach out to the clients proactively and provide information without them actively demanding it. Thus, adding real-time functionality to the Web applications was always a challenge. With new real-time

requirements, new technologies are emerging which give us many ways to push data from the server to connected clients as it happens in real-time. The real-time Web functionality, i.e., server side code pushing the contents to connected clients, has become essential as this technology platform enables communication across the clients, and clients don't need to poll the server. Thus real-time information push technology is a game changer that allows the development of more efficient applications with improved technology. Hence, we can monitor desktop applications with mobile applications in real-time.

ASP.NET SignalR is an asynchronous signalling library for ASP.NET developers, to help build multi-user interactive Web applications and make real-time functionality available to the applications. SignalR allows bi-directional communication between the client and the server. The server can push contents to the client once it becomes available. SignalR makes it extremely easy to add real-time connectivity to your applications, so that servers and their clients can push data back and forth to each other in real-time.

How SignalR works

SignalR is a set of server and client libraries that facilitate simple, real-time, two-way communications between server and clients. It is an open source API, and is accessible through GitHub. For documentation, refer to <https://github.com/SignalR/SignalR/wiki>. The SignalR library can be installed by going through the following steps:

1. Go to **Tools > Library Package Manager > Package**